2 The Chemical Context of Life

KEY CONCEPTS

Study Tip

Make a table: As you read the chapter, make a summary table like the following. Add more rows as you proceed.

Property	Element (atom) C	H	O	N
Atomic number				
# Electrons				
# Neutrons				
Mass number				
Electron distribution diagram				
# Valence electrons				

Go to Mastering Biology

For Students (in eText and Study Area)
- Get Ready for Chapter 2
- Figure 2.6 Walkthrough: Energy Levels of an Atom's Electrons
- Animation: Introduction to Chemical Bonds

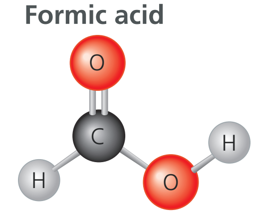
For Instructors to Assign (in Item Library)
- Chemistry Review: Atoms and Molecules: Covalent Bonds
- Chemistry Review: Atoms and Molecules: Electronegativity

Figure 2.1 Wood ants (*Formica rufa*) use chemistry to ward off enemies. When threatened from above, they shoot volleys of formic acid from their abdomens into the air. The acid bombards and stings potential predators, such as hungry birds.

What determines the properties of a compound such as formic acid?

A compound is made of atoms joined by bonds. Formic acid (CH_2O_2) consists of carbon (C), hydrogen (H), and oxygen (O).

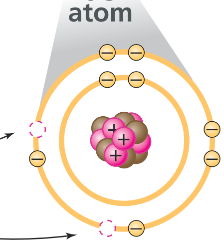
Formic acid



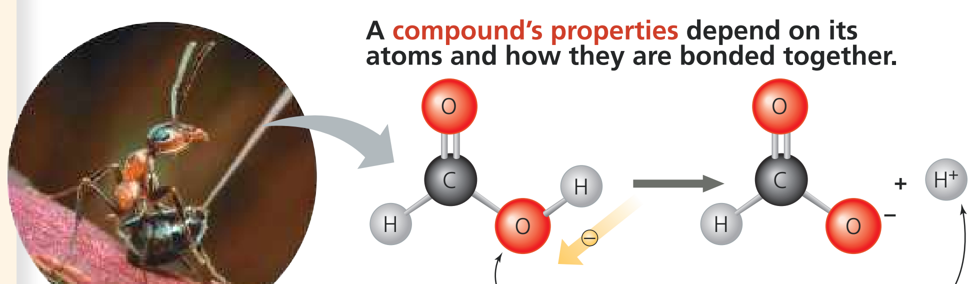
The number of protons (⊕) determines an atom's identity. Oxygen has 8 protons.

Oxygen atom

An atom's electron (⊖) distribution determines its ability to form bonds. Oxygen has space for 2 more electrons, so it can form 2 bonds.



A compound's properties depend on its atoms and how they are bonded together.



In formic acid, this O attracts H's electron, releasing H^+ and making this compound an acid, which stings.